

NEYMAN-PEARSON LEMMA AND ITS APPLICATIONS IN MACHINE LEARNING

PROJECT REPORT

SERIAL NUMBER: CMS/UGSIP26/043

**SUBMITTED FOR THE PARTIAL FULFILLMENT OF THE AWARD OF
INTERNSHIP PROGRAMME IN MATHEMATICS – 2026
CALCUTTA MATHEMATICAL SOCIETY**

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This is to certify that the project report entitled “**Neyman-Pearson Lemma and Its Applications in Machine Learning**” is submitted by **PRIYANSHU KUMAR** for the partial fulfillment of the award of **Internship Programme in Mathematics – 2026**, Calcutta Mathematical Society, under the sole supervision of **Dr. Kakali Karmakar (Sur)**, Assistant Professor, Department of Engineering Mathematics, B.P. Poddar Institute of Management and Technology, *Life Member of Calcutta Mathematical Society*. The report has been prepared in accordance with the prescribed format and is suitable for submission.

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Dr. Kakali Karmakar (Sur)**, Assistant Professor, Department of Engineering Mathematics, B.P. Poddar Institute of Management and Technology, for her invaluable guidance, constant support, and encouragement throughout the course of this internship project. Her insights and suggestions helped me shape the report in a clearer and more meaningful way.

I also wish to extend my thanks to the **Calcutta Mathematical Society** for offering me the opportunity to be a part of this internship programme. The exposure and experience I have gained through this project have been truly valuable for my academic growth.

I am grateful to **Indian Institute of Technology Patna** for providing the academic environment and support needed to complete this work. Lastly, I would like to thank my family and friends for their constant encouragement throughout this journey.

Priyanshu Kumar

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Internship Programme in Mathematics – 2026



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1 Introduction

Decision making under uncertainty is common in statistics and machine learning. Many practical problems are not satisfied by ordinary accuracy alone, because one kind of mistake is more costly than the other. A false alarm may be acceptable in a few cases, but missing an important case may be far more serious. The Neyman-Pearson lemma gives a clean way to handle this kind of asymmetry. Its basic message is that, when a limit is fixed on one error, the best test should be built by comparing how likely the observed data are under the two competing hypotheses. That idea is simple, but it has stayed useful for a long time because it matches real decision problems very well.

The same logic appears naturally in machine learning. A classifier also has to decide between two classes, and in many settings the user already knows which mistake should be controlled more tightly. This report gives a short account of the Neyman-Pearson idea, explains how it connects with binary classification, and shows why it remains useful in applications such as medical screening, fraud detection, anomaly detection, and other high-stakes prediction tasks. The aim is not to be mathematically heavy, but to present the idea clearly in a form that is easy to read and easy to present.

The report is organized in a simple way. First, the core statistical idea is introduced in plain language. Then the connection with machine learning is discussed through threshold-based classification and a few important research directions. After that, the report looks at applications and practical strengths. This structure keeps the discussion compact while still making the main ideas visible to a reader who may be seeing the topic for the first time.

For the basic background on hypothesis testing, likelihood ratio ideas, and the standard language of mathematical statistics, the report also follows the treatment found in [2]. That textbook helps keep the presentation aligned with the usual undergraduate and postgraduate style of statistical writing.

2 Discussion

2.1 Core idea

The Neyman-Pearson lemma is usually stated for two simple hypotheses,

$$\square_0 : \square \sim \square_0 \quad \text{and} \quad \square_1 : \square \sim \square_1.$$

The idea is to keep the type I error below a chosen level and then make the test as powerful as possible under that restriction. In practical terms, the decision is based on a likelihood ratio, which compares the support of the observed data under the two models. If the ratio is large, the data point looks more consistent with the alternative hypothesis; if it is small, the null hypothesis is preferred.

This rule is valuable because it turns a difficult decision problem into a simple threshold problem. Instead of searching for an arbitrary decision boundary, one chooses the boundary in a way that respects the required error control. That is why the lemma is still cited so often in statistical theory and in applied machine learning [1].

Theorem 1 (Neyman-Pearson Lemma). Let $\square_0$ and $\square_1$ be probability distributions with densities

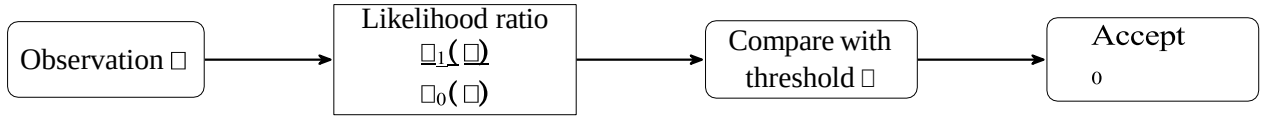


Figure 1: Basic Neyman-Pearson decision flow from score or likelihood information to a final thresholded decision.

l_0 and l_1 with respect to a common measure μ . Consider testing

$$H_0 : \mu_0 \quad \text{against} \quad H_1 : \mu_1.$$

If a test ϕ satisfies

$$\int \phi(x) d\mu_0(x) = \alpha,$$

and

$$\phi(x) = \begin{cases} 1, & \text{when } l_1(x) > c l_0(x), \\ 0, & \text{when } l_1(x) < c l_0(x), \end{cases}$$

for some constant c , then ϕ is the most powerful test of level α . Conversely, a most powerful test of level α must satisfy this likelihood-ratio form except possibly on a boundary set.

2.2 Historical note

The original result was developed in classical mathematical statistics, long before modern machine learning became popular. Even so, the same reasoning still appears in present-day classification problems because the basic issue has not changed: one often needs to decide between two possibilities while controlling the risk of a particular kind of error. That continuity is one reason the Neyman-Pearson lemma remains a standard topic in both theory and applications.

2.3 Connection with machine learning

In machine learning, binary classification is the closest analogue of this setup. A model often produces a score, and a final label is obtained by thresholding that score. The Neyman-Pearson viewpoint becomes useful when the two classes are not equally important. For example, if the positive class corresponds to a disease or a fraud case, then a missed positive case may be more damaging than a false alarm.

Several later papers developed this idea into practical classification methods. Plug-in rules showed that ordinary scoring models can be adapted to Neyman-Pearson constraints by calibrating the threshold carefully [7]. Sample-splitting ideas and the umbrella algorithm made the method easier to use with common classifiers [10]. More recent work has shown that the same framework can still work in high-dimensional settings, with missing data, and even under label noise [9, 12, 13]. This makes the approach relevant for real data rather than only for textbook examples.

2.4 Early learning perspective

Before the recent wave of machine learning papers, the Neyman-Pearson idea had already been studied from the learning-theory point of view. Scott and Nowak looked at constrained

versions of empirical risk minimization and structural risk minimization, showing that NP-style classification could be developed using the same kind of thinking that is used in statistical learning theory [4]. That work is important because it explains why the idea is not only a modification of classical testing, but also a genuine learning problem.

Another helpful contribution came from Scott's discussion of how to compare and design classifiers in the NP setting [5]. The paper emphasizes that there is no single natural performance measure when one error is constrained and the other is minimized. That point is useful even in a short student report, because it explains why NP classification is not judged in exactly the same way as ordinary classification.

Taken together, these early studies show that the Neyman-Pearson framework was never only about a single theorem. It gradually became a learning paradigm with its own evaluation questions, its own training strategy, and its own practical uses.

2.5 Applications

Medical diagnosis is one of the clearest applications. In screening problems, the goal is often to avoid missing a serious case, even if that means allowing a few false alarms. Fraud detection and cybersecurity have a similar structure, because letting a fraudulent action pass can be more harmful than checking one extra genuine transaction. Anomaly detection also fits naturally, since the rare event is what needs to be found without producing too many false alerts. Spam filtering and content moderation use the same logic when one type of error is more costly than the other.

The broader literature shows that this is not just a theoretical point. Early novelty-detection work already used Neyman-Pearson ideas to control false positives [6]. Later papers refined the theory, connected it with common machine learning models, and studied cases where the sample size is small or the feature dimension is large [3, 8, 11, 14]. These developments make the method useful in practice and not only in theory.

Related work on minimum volume sets and anomaly detectors also connects naturally to this framework [17, 18, 19]. These papers show that the same error-control mindset can support set estimation and highly imbalanced classification as well.

2.6 Genomics and biomarker studies

Another important area is genomics and biomarker discovery. In these problems, the number of features is often very large, but the number of truly relevant cases may be quite small. A learning rule that minimizes only overall error can miss the special class that matters most. The Neyman-Pearson formulation is attractive here because it lets the analyst focus on protecting the more important error while still making a useful prediction rule.

This line of work is especially natural in biomedical data analysis, where the consequence of overlooking a relevant signal can be serious. The genomic applications discussed by Li and Tong show that the NP viewpoint fits well in settings where the data are high-dimensional and the scientific cost of a mistake is asymmetric [15]. This adds another example of how the framework connects theory with practice.

2.7 A simple illustration

Consider a situation where a model produces a score for each observation, and higher scores indicate stronger evidence for the positive class. If the threshold is set too low, many normal observations will be incorrectly flagged. If the threshold is set too high, many positive cases will be missed. The Neyman-Pearson idea is to choose the threshold in a controlled way so that the false positive rate stays below a fixed level chosen in advance.

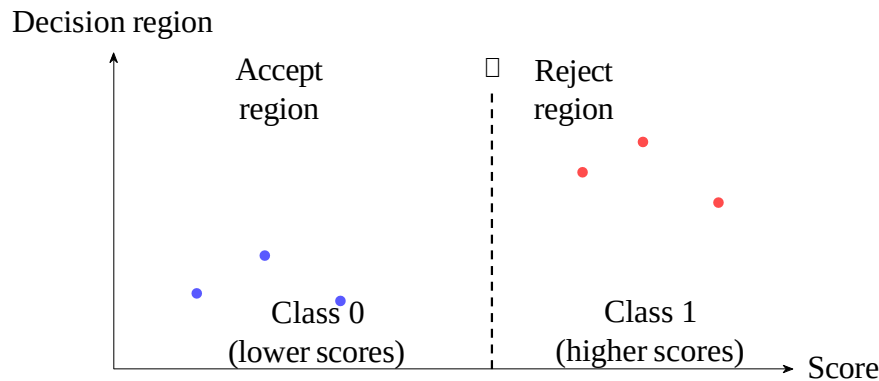


Figure 2: A simple threshold picture showing how a score is split into two decision regions.

This is exactly what makes the method attractive in practical work. The user does not have to rely on a vague balance between all mistakes together. Instead, the more serious error is treated first, and the remaining performance is optimized within that boundary. In a medical setting, this means the test can be tuned to avoid too many false alarms while still catching as many true cases as possible. In a fraud system, it means a bank can decide how much inconvenience it is willing to tolerate before the classifier starts blocking ordinary transactions.

Another useful point is that this threshold can often be explained to non-specialists. A doctor, engineer, or policy maker may not want a long theoretical description, but they can understand a statement such as: "We keep the false alarm rate below a chosen limit, and then we use the best remaining rule within that limit." That clarity is one reason the Neyman-Pearson framework stays relevant across different fields.

The same explanation also helps in classroom presentations and viva. If the examiner asks what the method really does, the student can answer in one line: it controls one error first and then improves the other as much as possible. That simple summary captures the spirit of the entire report without requiring heavy notation.

2.8 Strengths and limitations

The main strength of the Neyman-Pearson approach is that it makes the priority of errors explicit. This is often better than reporting only one overall accuracy number, because accuracy can hide the exact mistake that matters most. The method is also flexible, since it can be built on top of existing score-based classifiers instead of replacing them completely.

The main limitation is that strong error control can make the learning problem more delicate. Threshold calibration may need extra data, and guarantees are often easier to state than to implement perfectly in noisy data. Even so, the recent literature suggests that the framework is becoming more practical, especially through non-splitting methods and adjustments for label

noise [14, 13]. Overall, the Neyman-Pearson view gives a neat balance between theory and application.

2.9 Evaluation and metrics

In ordinary classification, accuracy is often the first number that people look at. For the Neyman-Pearson setting, however, accuracy is not enough on its own. A classifier may have a high overall score while still violating the error constraint on the protected class. That is why the main evaluation question changes from "How accurate is the model?" to "Does the model respect the required error bound, and how much power does it still retain within that bound?"

This change in emphasis is important for both theory and practice. It explains why NP classification papers often discuss type I error, type II error, power, and ROC-style curves rather than only overall accuracy. The score-based comparison is still useful, but it must be interpreted with the error constraint in mind. The NP-ROC idea and related evaluation tools were developed for exactly this reason [10]. They give a way to compare methods while keeping the control objective visible.

This also makes the results easier to compare between methods. Two approaches can have similar accuracy, but if only one of them respects the error bound consistently, then that one is better for the Neyman-Pearson problem. In that sense, the report should emphasize the controlled error as the main outcome and not treat accuracy as the whole story.

2.10 Operating points and NP-ROC curves

A particularly helpful way to summarize the performance of an NP classifier is to look at operating points. Each threshold corresponds to a different pair of type I and type II errors. If the threshold is moved slightly, the trade-off changes. This creates a curve rather than a single number. In the NP setting, that curve is read with the error constraint in mind. The goal is not to find the best average compromise, but to find a point where the constrained error is acceptable and the remaining error is as small as possible.

This point of view makes comparison easier. Two classifiers may have similar accuracy, but if one of them stays within the desired false-positive bound more reliably, then it is the better choice for the NP problem. That is why ROC-style summaries are useful in addition to ordinary performance tables. They show not only which model is stronger overall, but also which model is more stable at the chosen operating point [10].

2.11 Practical remark

In an actual project, it is often enough to explain the method at a high level rather than present every technical detail. A short report can say that the model is first trained to produce a score, and then a threshold is selected so that one chosen error stays within a fixed bound. This is a good summary because it connects the theorem with a useful working rule. It also helps during viva, since the examiner usually wants to see whether the student has understood the central idea and not only the formula.

For that reason, the report keeps the mathematics limited to the key likelihood-ratio statement and avoids unnecessary complication in the later sections. The aim is to make the project

readable, presentable, and easy to explain in a short discussion.

2.12 How the method is used in practice

In practice, an NP classifier is usually built in stages. First, a learning model is trained to produce a score or ranking. This score does not have to be a probability in a strict sense; it only needs to order observations from more suspicious to less suspicious. Second, the score is calibrated on a separate sample or validation set. Third, a threshold is chosen so that the specified type I error requirement is satisfied. Finally, the chosen rule is checked on new data to see how stable the control remains.

This sequence is important because it separates learning from decision control. If the same data are used carelessly for both steps, the error estimate may look better than it really is. That is why many NP methods use sample splitting or some form of independent threshold calibration [10, 11]. The recent non-splitting work is interesting precisely because it tries to reduce the loss of data caused by this separation while still keeping the error guarantee [14].

2.13 Relation to multi-class problems

Although the report focuses on the binary case, it is worth noting that recent work has started to extend NP ideas beyond two classes. This is not trivial, because in multi-class problems one must decide how to state the error priorities and how to check whether the constraints are feasible. A recent JASA paper connected the multi-class NP problem to cost-sensitive learning and proposed algorithms with theoretical guarantees [16]. This is a good sign that the NP philosophy is flexible enough to move beyond the simplest setting.

For a project report, this is a useful point to mention briefly rather than in full detail. It shows that the topic is still active and that the binary framework studied here is part of a larger research program. The student can therefore present the main lemma as the foundation while also showing awareness of current extensions.

2.14 Reading the result

When a practitioner looks at the output of an NP classifier, the first question is not simply whether the model is accurate overall. The real question is whether the protected error is actually under the chosen limit. If that condition holds, then the remaining error becomes the main quantity of interest. This is different from ordinary accuracy reporting, where all mistakes are mixed together. The NP viewpoint therefore gives a more transparent description of the classifier in settings where one class has to be treated with more care.

This kind of reading is very useful in reports and presentations. Instead of talking only about abstract risk or generic performance, the student can explain the result in a way that connects directly to the application. For example, in disease screening, the report can say that the system is designed to keep false alarms within a bound while still finding as many true cases as possible. In fraud detection, the same logic becomes a policy question about how many ordinary transactions can be reviewed without causing too much inconvenience.

3 Methodology

3.1 Problem formulation

The main problem considered in this report is binary classification under asymmetric error priorities. The data consist of observations that belong to one of two classes. A model is built to separate the classes, but the loss of one kind of mistake is treated as more severe than the other. This is the setting in which the Neyman-Pearson framework is most natural.

The first step is to choose which class deserves stronger protection. In many examples this is the positive class, but the framework itself does not depend on that choice. Once the protected error is fixed, the task becomes a constrained optimization problem. The model should respect the required error level and then do as well as possible on the remaining error. This simple formulation is one of the reasons the framework is easy to explain to non-specialists.

3.2 Learning a score

Most practical implementations begin with a score-based learner. The learner can be a logistic model, a naive Bayes rule, a discriminant rule, or any other classifier that produces a ranking. The exact model is less important than the fact that it gives a way to order observations according to how likely they are to belong to the positive class. Once such a score exists, the NP framework can be applied by choosing a suitable threshold.

This is one of the strengths of the approach. The user does not need to design a completely new learning system. Instead, a familiar machine learning model is used as the first stage, and the NP requirement is imposed at the thresholding stage. In that sense, the framework acts as a decision layer on top of a standard learner [7, 9].

3.3 Threshold choice

The threshold is the heart of the method. If it is too low, the classifier will make too many false positives. If it is too high, the classifier will miss too many true positives. The goal is to find a value that keeps the protected error below the chosen bound while preserving as much power as possible. This is usually done using a validation sample or a carefully designed calibration rule.

The threshold step is also where the theory becomes useful in practice. The Neyman-Pearson lemma tells us that a likelihood-ratio style rule is optimal when the distributions are known. In real data problems, the exact distributions are not known, so the learning task is to approximate that ideal rule as closely as possible. That is why the later literature focused on plug-in methods, oracle inequalities, and finite-sample guarantees [3, 7, 11].

3.4 Validation and reporting

After the rule is chosen, it must be checked on new data or on a holdout set. This check is important because a method that looks good during training may not keep the same error control later. In a real project, this is where the practical usefulness of the method becomes visible. If the false positive rate remains under the target level, the model has succeeded in the most important part of the task.

For a report, this stage should be explained clearly but without unnecessary technical detail. It is enough to say that the model is calibrated, tested, and then interpreted in terms of the controlled error. That keeps the discussion accessible while still showing that the student understands how the method is used in a real workflow.

3.5 Why sample splitting is used

Sample splitting appears often in the Neyman-Pearson literature because the same data should not be used too aggressively for every purpose. If one part of the sample is used to build the score and the same observations are also used to choose the threshold, the error control can become too optimistic. By separating the roles of the data, the method keeps the calibration step closer to the independent setting that the theory prefers.

This is one of the key practical ideas behind the umbrella algorithm and the related plug-in methods [10, 11]. It is also why recent research on non-splitting classifiers is interesting: it asks whether the same benefits can be achieved without sacrificing too much data. In small-sample problems this is especially important, because every observation matters. The method therefore has a clear statistical reason for using sample splitting, but it also has a clear practical reason for trying to reduce it when possible.

4 Illustrative Case Study

4.1 Medical screening scenario

Suppose a hospital is using a model to flag patients who may need additional screening. The model assigns each patient a score based on symptoms, test results, or other measurements. A higher score means stronger evidence that the patient belongs to the positive class. If the threshold is set too low, many healthy patients will be sent for unnecessary follow-up. If the threshold is set too high, some patients who actually need attention may be missed.

The Neyman-Pearson approach offers a clear policy for this kind of problem. The hospital first chooses an acceptable false alarm level. Then the score threshold is selected so that the false alarm rate stays below that level. Among all such thresholds, the one that gives the best remaining detection performance is preferred. This makes the choice easier to justify because the decision is based on a fixed safety requirement rather than on a vague compromise.

What is attractive about this setup is the clarity it gives to the user. Doctors do not need to think in terms of abstract optimization objectives. They can instead ask a simple operational question: how many unnecessary alerts are acceptable, and how strong should the model be within that limit? The Neyman-Pearson framework answers that question in a mathematically disciplined way.

4.2 Fraud detection scenario

A similar logic appears in financial fraud detection. A bank may tolerate some manual review of genuine transactions if that helps stop fraudulent ones early. But if the threshold is too strict, the system may interrupt too many legitimate payments and annoy customers. Here again, the

protected error is the false alarm rate, and the remaining objective is to catch as much fraud as possible within that limit.

This example shows why the NP framework is not only a theoretical curiosity. It is a way of translating business priorities into a model design choice. The threshold is not selected just because it looks good on a graph. It is selected because it satisfies a policy decision. That policy-based interpretation is one of the strongest reasons to study the lemma in a machine learning context.

4.3 What the example teaches

The broader lesson from these scenarios is that a classifier should be evaluated in the language of the application. In some projects, missing a positive case is expensive. In others, too many false alarms are the real problem. The Neyman-Pearson view makes this distinction explicit and builds it into the model. This is why the framework remains useful even though many modern learning methods are available.

For a student report, these examples are valuable because they make the theory easier to remember. A reader may forget the exact theorem statement, but they are likely to remember the hospital example or the fraud-detection example. That makes the report more effective both as a written submission and as preparation for a viva.

5 Future Directions

The recent literature suggests that the Neyman-Pearson framework still has room to grow. One important direction is the reduction of sample splitting, because using fewer observations for threshold calibration can make the method conservative. Another direction is robustness to label noise, since real data often come from human annotation or imperfect operational records. These two issues matter in practice because they affect both the statistical guarantee and the convenience of implementation.

High-dimensional data is another continuing challenge. In genomics, text analysis, and other modern applications, the number of predictors can be very large. The NP framework has already shown that it can work in this setting, but there is still room for improvement in efficiency and interpretability [9]. The missing-data literature is also important, because it shows how the framework can be extended when the data are incomplete rather than ideal [12].

Overall, the future of the topic seems to lie in methods that preserve the clear error-control philosophy while becoming easier to use on large, noisy, and complicated datasets. Recent work on distribution-free rates also points to a more general theory that does not depend on a narrow parametric assumption [20]. The recent move toward multi-class NP classification is especially important because many real-world problems are not naturally binary [16]. That direction is promising because it keeps the original statistical insight intact while making the framework more practical for modern machine learning problems.

6 Conclusion

The Neyman-Pearson lemma is important because it gives a principled way to make decisions when errors are not equally serious. That idea is useful in both statistics and machine learning, especially in binary classification problems where one class needs stronger protection than the other. The report showed how the classical lemma leads naturally to threshold-based decision rules and why those rules are helpful in practical settings such as medical diagnosis, fraud detection, and anomaly detection. The main takeaway is simple: when the cost of mistakes is unequal, a good model is not only one that predicts well, but one that controls the more serious error with care.

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